should be 1.0. The observation of deviations from 0.5 to 1.5 in (a) and up to 5 in (b) indicate that U, Th, Ra and Pb have migrated at different velocities in recent times. A detailed analysis yields the age of the U deposit and the migration rates of the daughter elements: the U and Th has not migrated during the past several hundred thousand years (this is probably also the age of the U deposit); Ra has migrated a few meters in the last 10 000 y.

13.8. AGE DETERMINATION FROM RADIOACTIVE DECAY

Prior to the discovery of radioactivity, geologists could obtain only poor estimates of the time scale of the evolution of the earth. The oldest geologic materials were assumed to be some 10 million years old, and it was believed that this represented the age of the earth. However, with the discovery of radioactivity early in this century, geologists developed more objective methods for such age determination ("nuclear clocks"). In 1907 B. B. Boltwood obtained a value of 2.2×10^9 y for the age uranium and thorium minerals, assuming that all U and Th ultimately decayed to lead. Considering how few isotopes in the chains had been discovered at the time, the calculation was surprisingly good. Nuclear clocks have provided primary data on the age and evolution of the earth (*nuclear geo-chronology*) as well as the formation of elements and of the universe (*cosmo-chronology*).

The cosmogenic radionuclides with relatively short half-lives can be used to date materials of more recent origin; e.g., ^{3}H for water movements in the geosphere, and ^{14}C for organic material of archaeologic interest. Practically all of the primordial radionuclides (Table 13.3) can be used for dating geologic materials: $^{40}K/^{40}Ar$ for igneous (plutonic) rocks (i.e. rocks which have solidified from a rather homogenous melt); $^{87}Rb/^{87}Sr$ for metamorphic and sedimentary rocks; $^{147}Sm/^{143}Nd$ for rock-forming silicate, phosphate and carbonate minerals; $^{187}Re/^{187}Os$ for sulfides and metallic material like iron meteorites, etc.

13.8.1. Dating by the ^{14}C method

It is reasonable to assume that the production of ^{14}C in the atmosphere has been rather constant for at least a million years, which means that equilibrium exists between the rates of formation and decay of the ^{14}C in the atmosphere. Moreover, the half-life of ^{14}C is sufficient to allow equilibrium between the ^{14}C in the atmosphere, the oceans (including precipitations to ocean bottoms), and exchangeable carbon in natural materials. Thus from measurement of the specific radioactivity of carbon, it should be possible to determine when the sample became isolated from its natural environmental compartment.

The discovery that all living organic material has a certain specific radioactivity due to ^{14}C led W. Libby to a new method for determination of the age of biological material. This method, which has been of great importance in dating archeological, geological,

etc, materials, is based on the assumption (i) that cosmogenic ^{14}C has been produced at a constant rate, (ii) that the amount of anthropologic ^{14}C is negligible compared to the cosmogenic, (iii) that after the organism incorporating the biological material died, no exchange occurs between the carbon atoms of the material and those of the surroundings. In such material the number of ^{14}C atoms decreases with time according to the half-life of ^{14}C. The equation is (for notation, see §5.13)

$$^{14}C\,(\mathrm{Bq}\,/\mathrm{g}) \;=\; {}^{14}C_{\mathrm{o}}\,\mathrm{e}^{-0.693t/5568} \tag{13.5a}$$

or

$$t\,(\mathrm{y}) \;=\; (\log {}^{14}C_{\mathrm{o}} - \log {}^{14}C)5568/0.301 \tag{13.5b}$$

where $^{14}C_{\mathrm{o}}$ is the initial ^{14}C activity of a standard ($^{14}C_{\mathrm{o}} \approx 14$ dpm/g, c.f. §13.4.3) at time of death of the plant, etc ($t = 0$). The reference time for ^{14}C ages is AD 1950, which is indicated with the letters bp or BP (for "before present"). The half-life of 5568 y is a standard reference value introduced by Libby. For example, if the specific activity of a sample is measured to be 0.1 dpm/g, then (13.5b) gives a value of 39 700 years as the time since the material ceased to exchange its carbon. Only with extreme care and very sophisticated equipment can a specimen very old be determined with reliability, but shorter times can be measured more accurately since the specific activities are larger. ^{14}C ages are used for dating specimen of 300 to 50 000 y, with an uncertainty of 10 − 100 y. ^{14}C-determinations were originally made by transferring the carbon into carbon dioxide, which was measured in an internal GM-counter (Ch. 9). Later, transformation into methane was preferred, and the CH_4 counted in an internal proportional counter. The most sensitive technique today is to introduce the sample into the ion source of a tandem-van-der-Graaff accelerator and "count" the amount of ^{14}C-ions relative to ^{12}C-ions in the beam by mass-spectrometry.

There are many cautions that must be observed in the use of carbon dating. In addition to those mentioned above, there is the possibility of isotopic effects in metabolic processes. These could cause ^{14}C to be slightly depleted relative to ^{12}C due to chemical reactions in the biological material. Many plants prefer ^{12}C over ^{13}C and ^{14}C. To take this into account to get the correct "solar age" instead of a slightly misleading "^{14}C age", the isotopic depletion can be determined from measured deviations in the $^{12}C/^{13}C$ ratio. A "$\delta^{13}C$ correction" is introduced according to

$$^{14}C_{\mathrm{corr}} \;=\; {}^{14}C\,\{1 - 2(\delta^{13}C + 25)/1000\} \tag{13.6}$$

The value of $\delta^{13}C$ differs for various substances; e.g. −35 to −20 for terrestrial organic matter, −8 to −7 for atmospheric CO_2, etc. Figure 13.6 illustrates the use of various $\delta^{13}C$ values for biological matter. These corrections lead to considerable adjustments of conventional ^{14}C ages; e.g. for the period 7 000 to 2 000 years ago,

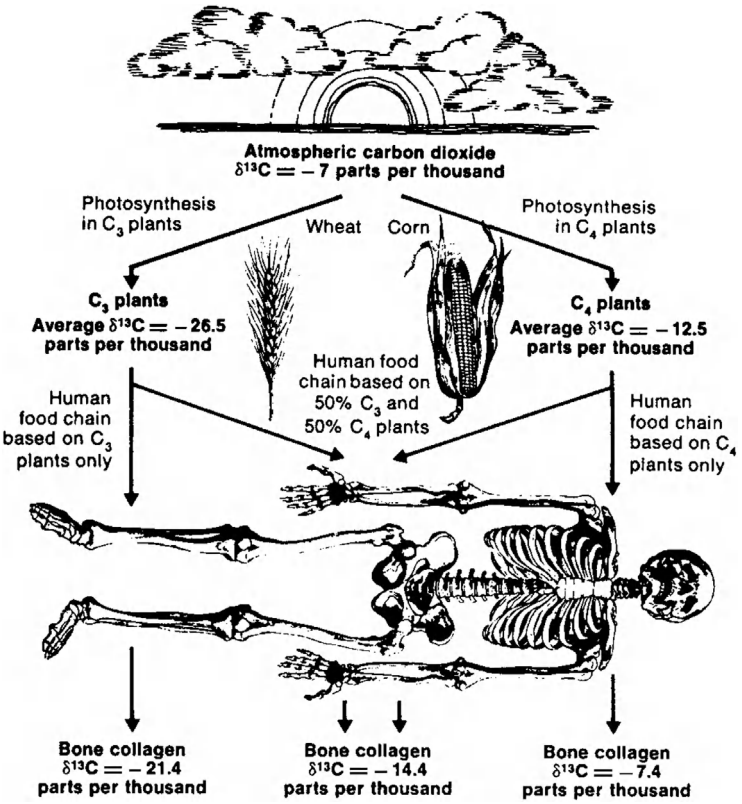


Figure 13.6 Ratio of $^{13}C/^{12}C$ in human bone depends on diet and affects the ^{13}C correction factor. *(From American Scientist, **70** (1982) 602).*

corrections increase ages by up to 1 000 y. The $\delta^{13}C$ can also be used to infer the relative size of the human contribution to atmospheric CO_2 by combustion of organics.

Another necessary correction has to be made for changes in the cosmic ray irradiation, which, *a priori* was assumed to be constant, but is not over long geological periods. By counting the number of annular rings on old trees such as the Sequoia gigantea, which can be almost 4 000 years old, and determining the ^{14}C content of each ring, the variation of cosmic radiation through thousands of years can be studied accurately, see Figure 13.7. Obviously there is a periodic cycle of some 10 000 y, which coincides with the variation in warm periods and small ice ages.

A large number of important age determinations have been made with the ^{14}C dating method. It was believed that a large ice cap had covered parts of the North American continent until about 35 000 years ago. Dating of wood and peat by the ^{14}C method has shown that the ice must have lasted until about 11 000 years ago. Moreover, several hundred pairs of sandals found in a cave in Oregon have been shown to be about 9000

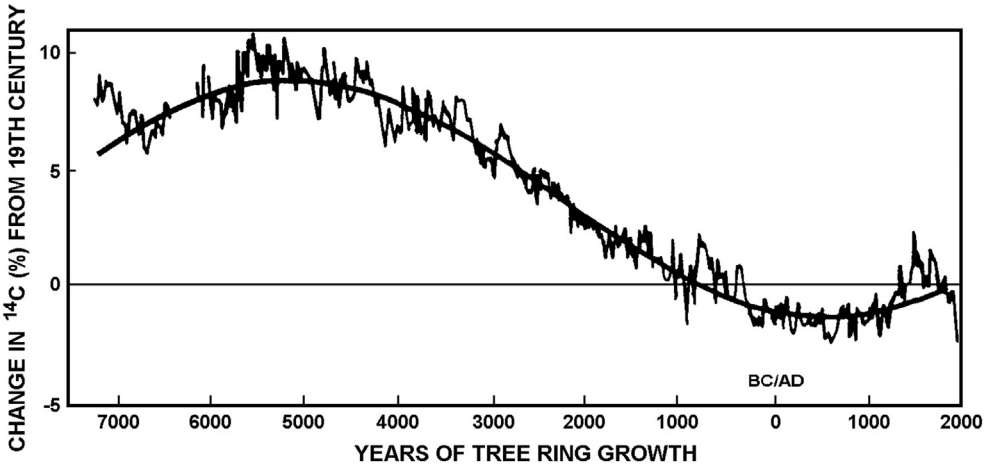


Figure 13.7 Long term variation in atmospheric ^{14}C activity as determined from tree rings. *(From H. E. Suess, La Jolla Radiocarbon Lab).*

years old, indicating that tribes with significant cultures had developed soon after the withdrawal of the ice cap from North America. By analyzing inorganic carbon in bone apatite, it has been determined that hunters in Arizona killed mammoths there 11 300 years ago. Another example of the use of ^{14}C dating, which has attracted widespread attention, involves the Dead Sea scrolls. There was considerable controversy about their authenticity until ^{14}C dating showed their age to be slightly more than 1900 years.

Some researchers claim that ^{14}C-ages lag behind with as much as 3500 years for times close to 20 000 y, when compared to U—Th dating (§13.8.4). Such adjustments play a large role in the debate about cultural migrations. It is interesting to note that adjusted ^{14}C-dates now have led to the conclusion that e.g. the stone monuments in Carnac, Brittany, are believed to be >6000 years old, i.e. older than the cultures in Egypt and Babylon. Similarly, grave mounds near the ancient edge of the inland ice in southern Norway and in Newfoundland have been found to be >7000 years old.

13.8.2. Dating by K-Ar method

Potassium is the eighth most abundant element in the earth's crust and occurs in many important rock-forming minerals. The radioactive isotope ^{40}K is present only to 0.0117% in natural potassium. It has a branched decay as follows:

$$^{40}\mathrm{K}\left(t_{½}\,1.28\times10^{9}\,\mathrm{y}\right)\begin{cases}\xrightarrow{\text{EC, }10.7\%}{}^{40}\mathrm{Ar}\ \ (99.6\%\text{ of natural Ar})\\ \xrightarrow[\beta^-,\ 89.3\%]{}{}^{40}\mathrm{Ca}\ \ (96.8\%\text{ of natural Ca})\end{cases}\qquad(13.7)$$

$\lambda_{EC} = 0.578 \times 10^{-10}$ y^{-1}, $\lambda_{\beta} = 4.837 \times 10^{-10}$ y^{-1}, E_{max} 1.32 MeV. Because the ^{40}K half-life is of the same magnitude as the age of the earth, measurement of the $^{40}K/Ar$ ratio can be used to determine ages of the oldest K-containing minerals (for notation, see §3.3.2). From the decay scheme one can derive the equation

$$t = \lambda^{-1} \ln[\{^{40}Ar/(0.107\,^{40}K)\} + 1] \tag{13.8}$$

where λ is the total decay constant ($= \lambda_{EC} + \lambda_{\beta}$; see (5.72)), ^{40}Ar represents the concentration of ^{40}Ar-atoms in the sample (i.e. number of radiogenic atoms of ^{40}Ar per unit sample weight), and ^{40}K is the present atomic abundance of ^{40}K-atoms. This equation assumes that all ^{40}Ar in the mineral is radiogenic. Any amount of non-radiogenic ^{40}Ar present initially (e.g. dissolved in the magma from which the mineral formed) is denoted as $^{40}Ar_i$. Thus the present amount of ^{40}Ar is the sum of the radiogenic amount and that originally present:

$$^{40}Ar = {}^{40}Ar_i + (\lambda_{EC}/\lambda)\{^{40}K(e^{\lambda t} - 1)\} \tag{13.9}$$

The problem is that both t and $^{40}Ar_i$ are unknown. This can be solved by dividing equation (13.9) by the number of ^{36}Ar atoms in the sample, yielding

$$^{40}Ar/^{36}Ar = (^{40}Ar/^{36}Ar)_i + (\lambda_{EC}/\lambda)(^{40}K/^{36}Ar)\{e^{\lambda t} - 1\} \tag{13.10}$$

Because ^{36}Ar is a stable non-radiogenic isotope, its amount in the sample should not change by time. Thus the measured $^{40}Ar/^{36}Ar$ ratio is the sum of the original and the radiogenic contributions from ^{40}Ar. Mineral samples of the same origin (so-called *cogenetic*) should have the same initial $^{40}Ar/^{36}Ar$ ratio and be of the same age (same t), though the $^{40}K/^{36}Ar$ may vary. Thus for cogenetic samples a plot of $^{40}Ar/^{36}Ar$ versus $^{40}K/^{36}Ar$ should yield a straight line, from the slope of which t is calculated. This line is referred to as an *isochron*. From the isochron for the rock sample from Tanzania in Figure 13.8, its age is calculated to be $(2.04 \pm 0.02) \times 10^6$ y, which is an interesting result as remains of early humans has been found in the same tuff matrix.

13.8.3. Dating by Rb–Sr method

Some uncertainty is associated with the use of the $^{40}K/^{40}Ar$ ratio because of the possibility of the loss of gaseous argon from minerals. An alternative method is based on ^{87}Rb/Sr system:

$$^{87}\text{Rb} \xrightarrow[4.8 \times 10^{10}\,\text{y}]{\beta^-} {}^{87}\text{Sr}\ (7.00\%\ \text{of natural Sr}) \tag{13.11}$$

The measurement of the $^{87}Rb/^{87}Sr$ ratio by isotope dilution and mass spectrometry is one of the most reliable methods for geologic age determinations. Meteorite values as

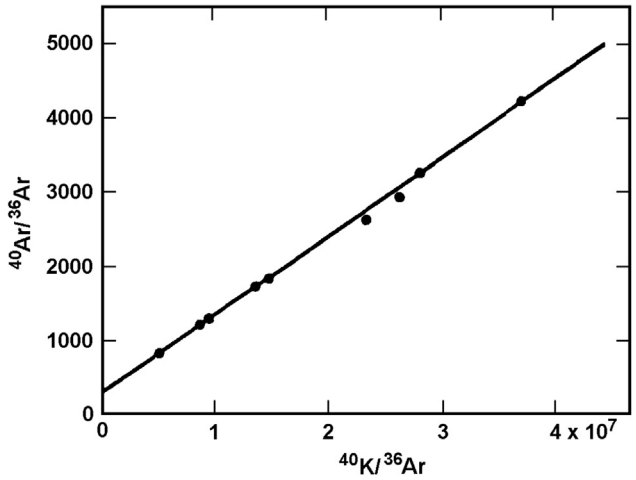


Figure 13.8 K-Ar isochron of tuff from Olduvai Gorge, Tanzania, yielding a slope corresponding to an age of 2.04 ± 0.02 My. *(From Faure).*

high as 4.7×10^9 y have been obtained, indicating that this is the age of the solar system. For the Rb–Sr clock a relation of type (13.11) is valid although corrections must be made for any non-radiogenic ^{87}Sr present; it is believed that the primordial $^{87}Sr/^{86}Sr$ ratio was 0.70.

13.8.4. Dating by the Re–Os method

The measurement of ^{187}Re, ^{186}Os, and ^{187}Os has been used to measure the age of meteorites and this yields some data about the age of our Milky Way galaxy.

$$^{187}\mathrm{Re} \xrightarrow[4.6 \times 10^{10}\,\mathrm{y}]{\beta^-} {}^{187}\mathrm{Os} \tag{13.12}$$

The insert in Figure 12.9 shows a part of the nuclide chart where the zigzag arrow shows nuclides formed in the s-process and the lower right arrows show nuclides formed in the r-process (c.f. §12.5.3). ^{186}Os can only form in the s-process and ^{187}Re only in the r-process. However, ^{187}Os is also formed by slow n-capture in ^{186}Os. The amount of ^{187}Os formed in the s-process can be calculated from estimated flux values and known cross sections (at 30 keV; 3.6×10^8 K). Thus the amount of ^{187}Os formed by decay of ^{187}Re can be calculated. This Re–Os clock gives us an age of 10.8 ± 2.2 eons from the formation of our Galaxy until our solar system formed. When combined with the age of the solar system, 4.7 eons, it yields an age of 15.5 ± 2.3 eons for our Galaxy, which seems to be too long when compared with the current best estimate of the time since the Big Bang, 13.75 ± 0.11 eons.

13.8.5. Dating based on ^{238}U decay

In the uranium decay series 8 α-particles are emitted in the decay from ^{238}U to ^{206}Pb. Thus for every 8 helium atoms found in a uranium mineral, one atom of ^{238}U must have decayed to ^{206}Pb. Designating the number of original uranium atoms in the sample at time 0 as $^{238}U_o$, the number which has decayed with time t would be $^{238}U_o - {}^{238}U$, where the last number refers to the uranium atoms present now. Then $^{238}U_o - {}^{238}U = He/8$, where He is the number of helium atoms made. From this it relatively easy to derive an equation from which t, i.e. the age of the mineral, can be calculated once the values of ^{238}U and He are known. For accurate calculations, it is necessary to correct for the formation of helium from the decay of the ^{235}U and ^{232}Th also present in the mineral (see Fig. 13.4). Further, if the mineral has lost any helium through diffusion or other processes during its existence, the helium content would be abnormally low, leading to erroneously small values of t. This method, therefore, can only give lower limits of the ages of minerals.

Another common method of dating U-minerals is by considering its content of lead isotopes. Lead has four stable isotopes of which three are end products of radioactive decay series. The fourth lead isotope, ^{204}Pb, is found in lead minerals in about 1.4% isotopic abundance and has no radio-genetic origin. At the time of formation of the earth, all the ^{204}Pb in nature must have been mixed with unknown amounts of the other lead isotopes. If a lead-containing mineral lacks ^{204}Pb, it can be assumed that presence of the other lead isotopes together with uranium and/or thorium must be due to their formation in the decay series. In such ^{204}Pb-free minerals if it is possible to determine the amount of the parent nuclide ^{238}U and of the end product ^{206}Pb, the age of the mineral can be obtained from the general equation

$$t = \lambda^{-1} \ln\left(1 + N_d/N_p\right) \tag{13.13}$$

where N_p is the number of parent atoms (e.g. ^{238}U) and N_d the number of radiogenic daughter atoms (e.g. ^{206}Pb), and λ is the decay constant of the parent. The atomic ratios between a number of isotopic pairs as a function of time is shown in Figure 13.9.

This method is more reliable than the helium method since there is very little possibility of any of the lead that has been formed by radioactive decay having diffused or been leached from the mineral during its geologic age. Mineral samples from the earth have yielded values as great as 3×10^9 y by the lead content method. Unfortunately, there is a problem with this method also. The decay series all pass through isotopes of the inert gas radon, and if some of the radon is lost from the mineral the ^{206}Pb content leads to an erroneously low age. However, if relation (13.13) is used for different decay series (i.e. the ^{238}U and the ^{235}U series) and the same t is obtained, the data are said to be *concordant*. Such data increases the confidence in the measured age.

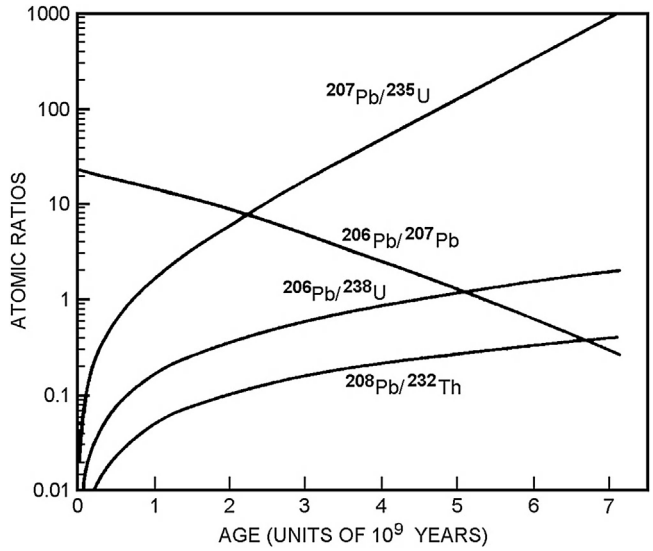


Figure 13.9 Atomic ratios at the time of measurement as a function of the age of the minerals. *(From E. K. Hyde).*

Since the lifetimes of the uranium isotopes ^{238}U and ^{235}U are different, the isotopic ratio between their end products ^{206}Pb and ^{207}Pb can also be used for age determination. One can derive the relationship

$$^{207}Pb/^{206}Pb = (1/138)\left(e^{\lambda_{235}t} - 1\right)/\left(e^{\lambda_{238}t} - 1\right) \tag{13.14}$$

where the factor 1/138 is the present isotopic abundance ratio of the uranium isotopes. This method has given values of 2.6×10^9 y for uranium and thorium minerals. When applied to stony meteorites, a somewhat longer age of $(4.55 \pm 0.07) \times 10^9$ y is obtained.

When lead is extracted from the ore it is in secular equilibrium with its precursors radium and uranium. The radium and most of its descendants are removed during processing while the ^{210}Pb accompanies the other lead isotopes. The separation of radium from lead is not always complete. Because of the long half-life of ^{226}Ra (1600 y) as compared to the short- lived ^{210}Pb (22.3 y), the determination of the excess ^{210}Pb over the equilibrium amount received from ^{226}Ra decay provides a scale for the time since the manufacture of the lead. This was used to verify forgeries of paintings claimed to be made by the Dutch artist Vermeer van Delft (1632–1675). The paintings were in fact made by H van Meegeren (1889–1948), in the 1940's and were so excellent in style that every authority accepted them as authentic. When threatened by the death penalty for selling such national treasures to the Germans during World War II, van Meegeren confessed the forgery, and his story was verified by dating the lead in the "lead white", $PbSO_4$, in the fake "Vermeers".

13.9. NATURAL RADIOACTIVITY OF THE OCEANS

The total amounts of ^{238}U and ^{232}Th in the ocean are 4.3×10^{12} kg (53 EBq) and 6.9×10^{10} kg (0.3 EBq), respectively. The oceans contain a much higher concentration of uranium than would be expected from its abundance in rocks. It also occurs at much larger concentration than thorium, even though the latter element is more abundant in rocks. This is explained by the absence of easily soluble thorium complexes, in contrast to the strong $UO_2(CO_3)_3^{4-}$ complex. ^{238}U decays to ^{234}Th ($t_{½}$ 24.1 d), which precipitates out or absorbs to colloids before it decays to ^{234}U, causing a slight disequilibrium between the two U-isotopes. This is used to determine the mean residence time of Th in the oceans: ~200 y; by contrast, the residence time of U in the ocean is 5×10^5 y. The decay passes the long-lived ^{230}Th ($t_{½}$ 75 400 y), which deposits in sediments before radioactive equilibrium is reached with ^{226}Ra (t 1600 y); consequently the specific activity of ^{226}Ra is less than expected from the U-content (39 Bq/m^3), only 1–10 Bq/m^3. It should be noted that the ^{222}Rn concentration in surface waters is in disequilibrium with ^{226}Ra because Rn is vented by heat, turbulence, etc.

As the total content of ^{40}K in the oceans is 7.4×10^{13} kg, corresponding to 1.94×10^4 EBq, ^{40}K is the largest source of radioactivity in the oceans. Minor activities come from ^3H as HTO and from ^{14}C as dissolved CO_2 or HCO_3^-.

13.10. ANTHROPOGENIC RADIOACTIVITY IN NATURE

In the analysis of a sample for its content of natural radioactivity it is necessary today to consider the possibility that the sample has become contaminated by "non-natural" radioactivities, i.e. radionuclides added to by human activities (so-called *anthropogenic sources*). Nuclear weapons tests, nuclear satellites burnt-up in the atmosphere, and nuclear power accidents may release large amounts of activities, see Table 13.4. The nuclear power industry is permitted by health authorities to continually release small, controlled amounts of specified radionuclides into the atmosphere and into open waters, Tables 19.7 and 21.10. Effluents may also come from large radiotracer users (especially

Table 13.4 Events leading to large injections of radionuclides into the atmosphere

Source	Country	Time	Radioactivity (Bq)	Important nuclides
Hiroshima & Nagasaki	Japan	1945	4×10^{16}	Fiss.prod., Actinides
Atmospheric weapons tests	USA USSR	-1963	2×10^{20}	Fiss. prod., Actinides
Windscale	UK	1957	1×10^{15}	^{131}I
Chelyabinsk (Kysthym)	USSR	1957	8×10^{16}	Fiss. prod. ^{90}Sr, ^{137}Cs
Harrisburg	USA	1979	1×10^{12}	Noble gases, ^{131}I
Chernobyl	USSR	1986	2×10^{18}	^{137}Cs
Fukushima	Japan	2010	2×10^{18}	^{131}I, ^{137}Cs